

Informing VR Teleportation interaction design through a RtD approach

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This study explores the iterative design of VR teleportation mechanics through RtD, focusing on usability trade-offs between two interaction models. The first iteration distributed tasks across both hands: the right hand activated teleportation using the "A" button, while the left joystick controlled orientation, enhancing precision but limiting multitasking. The second iteration centralized interaction on the dominant hand by using the trigger button for activation and the joystick for orientation adjustment. This approach improved flexibility, accommodating left-handed users and individuals with limited mobility, but introduced a steeper learning curve and increased user strain. Further evaluation is needed to optimize the balance between efficiency and usability.

CCS Concepts: • **Human-centered computing** → **Scenario-based design**.

Additional Key Words and Phrases: VR Teleportation, Game Development, High-paced video games, stressor, HCI

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1 Introduction

VR locomotion techniques enable users to navigate virtual environments by controlling avatars without requiring physical movement. Among these methods, teleportation [3] has been extensively studied and compared against other techniques and various modifications [4, 8, 13]. Teleportation is particularly valued in VR games for its ability to minimize virtual sickness and enable rapid spatial traversal. Development platforms like the Meta Quest have incorporated the classic Point & Teleport method as their default locomotion feature, underscoring its practicality and simplicity. However, while much of the literature highlights new variations of teleportation and user evaluations, the design processes that shape these methods often remain unpublished. This gap leaves questions about how designers address challenges during prototyping and what insights arise from iterative experimentation.

To address this gap, this study documents and presents the iterative development of teleportation systems on the Meta Quest 3 VR headset, employing a research-through-design (RtD) [14, 15] approach. As part of a larger project investigating the effects of in-game challenges on teleportation techniques, in this generative phase, RtD served as a framework to explore how teleportation mechanics can be adapted and refined to tackle the following research questions: **RQ1: How does the iterative RtD approach inform the development of VR teleportation systems?** and **RQ2: What learnings can be drawn from teleportation design iterations to enhance future VR interaction techniques?**

By iteratively prototyping and reflecting on two designs—one distributing teleportation control across both hands and another centralizing interaction into a single hand—this study examines nuanced trade-offs in efficiency, accessibility, and learning curves for VR locomotion. The primary **contributions** of this work include the high-fidelity prototypes

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developed through iterative cycles, the insights gained from reflecting on their usability, and the broader implications for refining VR teleportation mechanics informed by these prototypes.

2 Background

In this section, we describe teleportation as a means of VR locomotion and the active interest of the research community, RtD, and the benefits of thinking through the generative process of prototyping, and present the current functionality of teleportation in modern development platforms such as the Meta Quest.

2.1 Teleportation in VR

Teleportation is a method of locomotion widely used in VR games. It is a standard feature in today's games, particularly in first-person shooters where accuracy and speed are critical, such as in *Hyper-Dash*¹ and *Doom VFR*². First introduced in 2016 as the Point & Teleport technique [3], it has been extensively explored and evaluated. This technique offered an effective solution for fast scene traversal without physical movement while reducing virtual sickness compared to other locomotion approaches, such as continuous movement controlled by a joystick or head tilting [7]. However, teleportation also introduced challenges, particularly with path integration and spatial orientation, as it creates an instant change in the rendered image in the player's viewport. This instantaneous image refresh has driven the exploration of viewport transitions, hyper-jumps, or dashes as alternatives to improve the player's experience [1, 2, 11]. Over the years, various modifications to the teleportation technique have been proposed and evaluated. These include changes to factors such as orientation at the destination, height adjustments, and 3D in-air teleportation [4, 8, 13]. These studies focus on balancing immersion and minimizing virtual sickness, showcasing the flexibility of teleportation as a locomotion technique. Despite the continued interest in understanding and enhancing teleportation mechanics, companies like Meta still adopt the classic Point & Teleport as the default locomotion feature in their SDK, reflecting its simplicity and effectiveness in practical applications.

2.2 Thinking through prototyping

Previous research has predominantly centered on introducing new teleportation techniques and their variations, often validated through user studies. However, limited attention has been paid to how iterative design processes can inform and refine these mechanics. Some caveats and nuances can only be discovered through generative exploration, which is in itself knowledge gained through the act of design (and prototyping). This generation of new knowledge informed by exploring the design space via the creation of artifacts is known as Research through Design.

RtD is a recognized research methodology in the HCI community [15], serving as a valuable tool in our work for exploring VR interactions. Within the RtD framework, the artifact itself—in this case, teleportation prototypes—becomes an implicit theoretical contribution. These prototypes codify the designer's understanding of the design space and interaction challenges, often revealing insights that would remain inaccessible without hands-on prototyping. Numerous examples in HCI research demonstrate how RtD fosters a practical and reflective approach to design exploration [5, 6, 10, 12]. RtD emphasizes the iterative development and refinement of prototypes as a method of inquiry. This process facilitates knowledge generation through cycles of making, testing, and reflecting, which proves particularly valuable in immersive VR contexts [9]. The dynamic and iterative nature of RtD allows researchers to immediately evaluate the effectiveness of their designs via self-testing and rapid prototyping, yielding actionable insights into user

¹https://store.steampowered.com/app/1386890/Hyper_Dash/

²https://store.steampowered.com/app/650000/DOOM_VFR/

interaction and system performance. By adopting RtD, this research delves into the design challenges associated with implementing teleportation mechanics in the Meta Quest 3, such as balancing usability, accessibility, and task efficiency. The iterative process has enabled the identification and resolution of potential usability issues early in the design cycle. This approach not only highlights the nuances of interaction design but also underscores the critical role of iterative development in refining VR locomotion systems.

2.3 VR Teleportation in practice

We chose the Meta Quest device as a prototyping platform because of its accessible software development kit (SDK), low price tag, and compatibility with the Unity game engine. This enabled us to implement ideas much faster. The device comes with one symmetrical controller for each hand with seven inputs. In their SDK, the default teleportation method is Point&Teleport, which implements teleportation using the joystick to initiate and trigger it. Developers can enable the user to teleport on a navigation surface (mesh) or define teleportation hotspots that specify a destination, height, and viewing orientation. The former was chosen for this study as it is more flexible and allows user exploration.

The default interaction has two states: **select**, which displays an arc from the controller to a teleportable area on the ground, and **activation**, which executes the actual teleportation, translating the player to the selected location instantly. Through the control, users activate the teleportation arc by slightly leaning the joystick forward, and by passing a defined threshold forward, the teleportation is triggered. This design is ambidextrous, supporting both hands and allows the teleportation to be canceled by moving the joystick sideways or pressing another button. The arc visualization displays both the teleport destination and the area to which the user is allowed to teleport. If the arc is cast on a surface that is not covered by the navigation mesh, the endpoint shows a cross (X) in red, and the arc is tinted reddish, indicating that it is not an allowed destination. If the user tries to trigger, the system outputs an error sound. Successfully activating teleportation on a destination also triggers a sound, an onomatopoeia of a swift movement.

2.3.1 Conceptual framework of orientation definition in teleportation. Orientation definition in teleportation allows users to pre-set their viewport orientation at the teleportation destination before their position is translated [3, 4]. Previous implementations include tracking the hand [3], using the controller’s built-in gyroscope or a trackpad [4] to adjust orientation. While effective in specific contexts, the implementation of VR interactions often relies heavily on platform-specific features and hardware, necessitating custom solutions for each device and limiting their adaptability. However, as VR hardware continues to evolve and standardization increases, these challenges may become less significant. In 2024, the controllers on the PlayStation VR2 ³, the HTC VIVE Focus 3 VR-Headset ⁴ and the Meta Quest 3 ⁵ have practically the same input system. With the Meta Quest 3’s controllers, the joystick emerged as a natural and intuitive mechanism for managing orientation. However, the default Meta teleportation system uses the joystick’s forward movement to trigger teleportation, while other directions cancel the action. This design conflicted with the need for simultaneous orientation adjustment, requiring a reconfiguration of the interaction system to integrate orientation control seamlessly. The subsequent sections detail the iterative process of designing the interaction system to accommodate the orientation using the joystick.

³<https://www.playstation.com/sv-se/ps-vr2/>

⁴<https://www.dustin.se/product/5011264058/vive-focus-3-vr-headset>

⁵<https://www.meta.com/se/quest/quest-3/>

3 Design study

To refine the teleportation interaction, we iteratively tested two designs, starting with a two-handed control system and evolving into a single-handed interaction model. The experimental software was developed on the Meta Quest 3 VR headset using Meta's Interaction SDK in conjunction with Unity's OpenXR SDK and its Built-in Render Pipeline. The application is installed directly onto the HMD for a wireless experience, and the viewport must be streamed to a MacBook Pro via the Meta Quest Developer Hub for real-time monitoring. More implementation details can be found at the GitHub repository⁶. The interaction design of Meta's default teleportation underwent several changes to support both Point, teleport, and Orientation definitions. Let us delve into the process.

3.1 First iteration

Two simultaneous actions needed to be supported to incorporate teleportation destination selection and orientation adjustment. Users first select a destination via the teleportation arc and then define the orientation at the arc's endpoint on the ground. Initially, a toggle or switch to hold the teleportation arc indefinitely until orientation is defined was considered, as it enables more control without having to hold on to an input. Ultimately, the interaction was rejected due to inefficiency compared to the fast and fluid default teleportation interaction. Inspired by the VR game Hyper-Dash⁷, where pressing the primary "A" button selects a destination and releasing it triggers teleportation, a button-based approach was adapted for the prototype's first iteration. However, this implementation occupied the right thumb, raising the question of how the joystick could effectively be used for orientation. To resolve this, we distributed the workload across both hands: the left hand adjusted orientation using the left control's joystick, while the right hand handled activation and teleportation triggering.

3.1.1 Functionality. A large 3D arrow, composed of a cylinder and a cone, indicates the user's desired orientation. This arrow appears at the teleportation arc's endpoint, defaulting to the user's current forward vector, which describes the direction in front. Orientation is adjusted by moving the left joystick using the thumb, which dynamically rotates the arrow. This system derives a 3D viewing orientation vector using the joystick's X and Y axes, which output a real floating-point number from $[-1.0, 1.0]$. The joystick's Y values are assigned to the orientation Z component, while the X component is set to zero, ensuring the user's vertical orientation remains unchanged. Once the user releases the joystick, the arrow locks in the defined direction. Releasing the right-hand "A" button then triggers teleportation to the defined destination and orientation.

During the process, we observed that separating teleportation tasks between hands could become problematic if users needed to use their left hand for tasks such as grabbing or holding virtual objects. This design emphasizes location selection, potentially causing usability challenges when the left hand is required for other interactions.

3.2 Second iteration

To address scenarios where the non-dominant hand is engaged in other tasks during teleportation, we revisited the interaction design with a new goal: enabling users to control both destination selection and orientation adjustment using a single hand. This iteration explored how to centralize teleportation controls on the dominant hand while maintaining the intuitive efficiency of the previous design.

⁶https://github.com/DavidGiraldoCode/s-Evaluating_VR_teleportation_under_stressful_gameplay

⁷https://store.steampowered.com/app/1386890/Hyper_Dash/

Insights from the first iteration provided essential guidance. The joystick proved effective and efficient for controlling orientation, necessitating the thumb's availability for this task. Additionally, destination selection required fast activation to ensure seamless gameplay. Considering these factors, we evaluated alternative inputs on the controller. Moving away from the Hyper-Dash reference, the trigger button—positioned on the front of the controller—emerged as a suitable option. The input system was subsequently remapped to utilize the trigger button for teleportation activation instead of the primary "A" button.

3.2.1 Functionality. In this design, the user presses the trigger button with their index finger to display the teleportation arc and select a destination. While the trigger is pressed, the arc dynamically updates as the user moves the controller, casting onto teleportation-enabled areas. Simultaneously, the user's thumb manipulates the joystick to adjust the orientation arrow, defining the viewport orientation at the teleportation destination.

4 Results

Through iterative design, we developed two-handed and single-handed teleportation models that aim to balance efficiency and usability. However, the single-handed model requires further testing, particularly with novice users, as it demands higher levels of motor coordination and precision. We elaborate further on these ideas and present a set of general improvements to the system.

The **context of use heavily influences the design of input systems**. For instance, in first-person shooter (FPS) games like Hyper Dash⁸, the trigger button is typically reserved for weapon activation. This association makes the single-handed teleportation model unsuitable for such contexts, necessitating alternative input strategies to align with gameplay conventions. Testing revealed that using the trigger button provided a firmer grip on the controller, which improved joystick precision. However, **the centralized single-hand interaction introduced a steeper learning curve and increased demands on finger coordination**. While experienced users may quickly adapt to this system, novice users might find the increased strain challenging. This highlights the need for further evaluation to determine the design's long-term impact on usability, comfort, and accessibility.

Lastly, we present general improvements to the system: orientation changes in the system are currently mapped one-to-one with joystick movement, which allows for quick adjustments but makes fine-tuning challenging. A smoother adjustment mechanism could improve user precision, enabling more subtle orientation control. Additionally, the current system lacks sufficient orientation feedback, particularly when the target is far away, where the large arrow indicator becomes less clear. Introducing an orientation indicator on the user's hand could help enhance visualization and provide more precise direction, improving user experience. Furthermore, audio feedback, which was absent in this prototype, is an important feature to consider in future iterations. Implementing sound cues during orientation adjustments would give users better feedback, enhancing their sense of control and providing more intuitive interaction with the system.

5 Discussion and Conclusion

We presented an iterative prototyping process by adopting a Research-through-Design (RtD) approach to explore and reflect on the design of VR teleportation interactions on the Meta Quest 3. This approach allowed us to identify how interaction design decisions, such as teleportation triggering and viewport orientation selection, impact user experience. Building such artifacts helped us reveal potential usability issues, analyze interaction trade-offs, and gain insights

⁸https://store.steampowered.com/app/1386890/Hyper_Dash/

into the design considerations required to enhance VR locomotion systems, such as balancing factors like workload distribution among controllers, joystick precision, learning curve, and accessibility.

5.1 Researching through prototyping to answer RQ1

During our research-through-design phase, we prioritized flexibility and iterative exploration over feature completeness, emphasizing rapid prototyping cycles to uncover usability challenges and design opportunities. We consider that RtD proved particularly effective for VR design as it exposed us to empirical data at an early stage, creating familiarity with the technology and experiencing firsthand what our design assumption entailed. We were unable to derive actionable insights that informed successive iterations by conducting self-testing and lab-based prototyping.

5.1.1 Understanding the tools to work with: High-fidelity prototyping requires technical expertise, and this study was no exception. Due to compatibility issues with MacOS and Unreal Engine, Unity was chosen as the development environment, simplifying the integration of VR features. Moreover, buffering time diligently can soften the impact of setting up development environments, which can involve navigating multiple dependencies. Unity’s VR ecosystem required almost a week of proper setup, including installing Android Bridge, Meta Quest Developer Hub, and a Vulkan-based VR virtual simulator. We are aware that technological literacy and programming skills may become a barrier for some practitioners in the design field. Still, today, new low-code tools such as ShapesXR⁹ are making research-through-design in VR more accessible by allowing fast VR prototyping.

5.2 Insights from Design Iterations to answer RQ2

The iterative design process revealed critical trade-offs in VR teleportation systems. In the first iteration, workload distribution assigned orientation control to the left joystick and activation to the right hand, effectively dividing tasks between hands. While this approach enhanced precision, it limited multitasking, particularly when the non-dominant hand was required for other tasks. The second iteration centralized controls on the dominant hand, utilizing the trigger button for activation and the joystick for orientation adjustment. This design offered greater flexibility and addressed the limitations of the two-handed setup. However, it introduced a steeper learning curve and increased strain during prolonged use, particularly for novice users. These findings highlight the need to balance usability and efficiency. While experienced users may adapt quickly to centralized controls, novice users could face increased cognitive and motor demands. Ensuring accessibility for beginners is critical, alongside offering advanced control options for experienced users. Further evaluation is necessary to assess the long-term impact of these trade-offs.

5.3 Conclusion and Future Directions

The iterative design process demonstrated the effectiveness of RtD in refining VR interaction systems and exploring usability challenges. Prototypes served as reflective tools, offering valuable insights into balancing workload, precision, and flexibility of use. However, these findings remain preliminary and highlight the need for future user studies to validate the designs and explore their applicability across diverse contexts. Future research should focus on user-centered evaluations to investigate how these insights translate to broader populations, particularly when the context has a more significant influence on the interaction system. By integrating these learnings into iterative prototyping cycles, VR designers can advance teleportation mechanics and other interaction paradigms, ensuring more usable and engaging virtual experiences for users.

⁹<https://www.shapesxr.com/>

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